A client's requirement is, he wants to predict the insurance charges based on the several parameters. The Client has provided the dataset of the same.

1.) Identify your problem statement

The client wants to **predict the insurance charges** of customers based on several parameters such as age, gender, BMI, number of children, and smoking habit.

This is a **Supervised Machine Learning – Regression Problem**, because:

- We already have historical data (dataset provided by the client)
- The target variable (**charges**) is continuous numeric data
- The goal is to build a model that predicts future insurance charges accurately.

2.) Tell basic info about the dataset (Total number of rows, columns)

Basic Information About the Dataset

Columns in dataset:

Column Name	Description	Type
age	Age of the person	Numerical
sex	Gender (male/female)	Categorical
bmi	Body Mass Index	Numerical
children	Number of children	Numerical
smoker	Whether the person smokes (yes/no)	Categorical
charges	Insurance charges	Numerical

Input(independent) – Age,Sex,Bmi,Children,Smoker

Output(target variable/dependent variable) - Charges

- Total Rows: **1338**
- Total Columns: **6**

3.) Mention the pre-processing method if you're doing any (like converting

- string to number – nominal data
  Yes we have converted categorical column Sex(male/female) and Smoker(yes/no) to Numerical.
- We converted as below by using One Hot Encoding
  sex_male = 0 or 1
  smoker_yes = 0 or 1
- For SVM we used Standardization as pre processing method
- **Train-Test Split**
  Dataset split into:
  - 80% Training Data
  - 20% Testing Data

4.) Develop a good model with r2_score. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.

We trained multiple regression models to find the best performing one.

Models used:

1. Linear Regression
2. Random Forest Regressor
3. Support Vector Regressor (SVR)
4. Decision Tree

Evaluation Metric: R2 Score/ R score

R^2 Score measures how well the model explains the variance of the target variable.

Range:

Best = 1
Average = 0 to 1
Poor = Negative

5.) All the research values (r2_score of the models) should be documented. (You can make tabulation or screenshot of the results.)

1. Linear Regression

Multi linear regression R2 score = 0.7894

Refer program in GITHUB - https://github.com/kschezhian25/WEEK-03-ML--ASSIGNMENT/blob/main/1.Linear%20Regression%20Model.ipynb

2.SVM (Support Vector Regression)

R2 Score value Table

S.no	***Hyper Parameter***	***gamma***	R Score
TC1	***linear***	***scale***	-0.0101
TC2	***rbf***	***scale***	-0.0833
TC3	***poly***	***scale***	-0.0756
TC4	***sigmoid***	***scale***	-0.0754
TC5	***precomputed***	***scale***	Not possible
TC6	***linear***	***auto***	-0.0101
TC7	***rbf***	***auto***	-0.0833
TC8	***poly***	***auto***	-0.0756
TC9	***sigmoid***	***auto***	-0.0833
TC10	***precomputed***	***auto***	Not possible

Kernal = Precomputed cannot be used for this dataset. For precomputed we need NXN matrix (Rows and columns should be equal. In this data set we have 1338X6.

SVM is -ve (less than 0)-it's a poor model for this data set

SVM R2 score = -0.0833

Code – GITHUB - https://github.com/kschezhian25/WEEK-03-ML--ASSIGNMENT/blob/main/2.SVM%20Model.ipynb

3.Decision Tree

S.no	CRITERION	SPLITTER	max_features	R Value
TC-1	*squared_error*	*best*	None	0.6896
TC-2	*squared_error*	*best*	***sqrt***	0.6654
TC-3	*squared_error*	*best*	***log2***	0.6408
TC-4	*friedman_mse*	*best*	None	0.7115
TC-5	*friedman_mse*	*best*	***sqrt***	0.6671
TC-6	*friedman_mse*	*best*	***log2***	0.7026
TC-7	*absolute_error*	*best*	None	0.6768
TC-8	*absolute_error*	*best*	***sqrt***	0.7382
TC-9	*absolute_error*	*best*	***log2***	0.6750
TC-10	*poisson*	*best*	None	0.7263
TC-11	*poisson*	*best*	***sqrt***	0.7129
TC-12	*poisson*	*best*	***log2***	0.7332
TC-13	*squared_error*	*random*	None	0.6608
TC-14	*squared_error*	*random*	***sqrt***	0.7394
TC-15	*squared_error*	*random*	***log2***	0.6006
TC-16	*friedman_mse*	*random*	None	0.7048
TC-17	*friedman_mse*	*random*	***sqrt***	0.6925
TC-18	*friedman_mse*	*random*	***log2***	0.6707
TC-19	***absolute_error***	***random***	**None**	**0.7410**
TC-20	*absolute_error*	*random*	***sqrt***	0.6524
TC-21	*absolute_error*	*random*	***log2***	0.6048
TC-22	*poisson*	*random*	None	0.7300
TC-23	*poisson*	*random*	***sqrt***	0.6962
TC-24	*poisson*	*random*	***log2***	0.6993

TC19 – absolute_error,random,None gives R2 score =0.7410

Code-GITHUB - https://github.com/kschezhian25/WEEK-03-ML--ASSIGNMENT/blob/main/3.Decision%20Tree%20Model.ipynb

4.Random Forest

s.no	***criterion***	**max_features**	***n_estimators***	R value
TC1	***squared_error***	***sqrt***	10	0.8454
TC2	***squared_error***	***sqrt***	100	0.8704
TC3	***squared_error***	***log2***	10	0.8480
TC4	***squared_error***	***log2***	100	0.8698
TC5	***squared_error***	***None***	10	0.8452
TC6	***squared_error***	***None***	100	0.8544
TC7	***absolute_error***	***sqrt***	10	0.8558
TC8	***absolute_error***	***sqrt***	100	0.8731
TC9	***absolute_error***	***log2***	10	0.8614
TC10	***absolute_error***	***log2***	100	0.8715
TC11	***absolute_error***	***None***	10	0.8370
TC12	***absolute_error***	***None***	100	0.8522
TC13	***friedman_mse***	***sqrt***	10	0.8612
TC14	***friedman_mse***	***sqrt***	100	0.8723
TC15	***friedman_mse***	***log2***	10	0.8517
TC16	***friedman_mse***	***log2***	100	0.8684
TC17	***friedman_mse***	***None***	10	0.8440
TC18	***friedman_mse***	***None***	100	0.8535
TC19	***poisson***	***sqrt***	10	0.8538
TC20	***poisson***	***sqrt***	100	0.8707
TC21	***poisson***	***log2***	10	0.8571
TC22	***poisson***	***log2***	100	0.8710
TC23	***poisson***	***None***	10	0.8314
TC24	***poisson***	***None***	100	0.8536

TC8 -R2 Score while using Random forest provide 0.8731 (85%)

Code -GITHUB - https://github.com/kschezhian25/WEEK-03-ML--ASSIGNMENT/blob/main/4.Random%20Forest.ipynb

6.) Mention your final model, justify why u have chosen the same.

Model	R2 Score	Comments
Linear Regression	**0.7894**	
Support Vector Regression	**-0.0833**	After standardization also the R2Score did not improve
Decision Tree	**0.7410**	
Random Forest Regressor	**0.8731**	

Final Model:

Random Forest Regressor

R2 Score: 0.8731

Why this model was selected

Random Forest was chosen because:

1. Highest R^2 score among all models.
2. Works well with mixed data types.
3. Reduces overfitting using multiple decision trees.
4. Provides stable predictions.